

# Worksheet · Section 3.8

## Iteration



### The Big Picture

Loops repeat code. Practice counting how many times a loop runs, tracing accumulators, fixing infinite loops, and writing your own loops.

### Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- |                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| _____ 1. <b>Iteration</b>      | <b>A.</b> Runs a block an exact number of times                   |
| _____ 2. <b>REPEAT n TIMES</b> | <b>B.</b> A Boolean expression that controls when a loop stops    |
| _____ 3. <b>REPEAT UNTIL</b>   | <b>C.</b> Runs a block once for every element of a list           |
| _____ 4. <b>FOR EACH</b>       | <b>D.</b> The block of code that repeats                          |
| _____ 5. <b>Infinite loop</b>  | <b>E.</b> A loop whose ending condition never becomes true        |
| _____ 6. <b>Counter</b>        | <b>F.</b> Repeating a set of steps                                |
| _____ 7. <b>Accumulator</b>    | <b>G.</b> Runs a block until a condition becomes true             |
| _____ 8. <b>Traversal</b>      | <b>H.</b> A variable that counts how many times something happens |
| _____ 9. <b>Loop body</b>      | <b>I.</b> A variable that builds up a total as a loop runs        |
| _____ 10. <b>Condition</b>     | <b>J.</b> Visiting the elements of a list one at a time           |

### Part 2 · Fill It In

Use each word from the word bank once.

1. REPEAT n TIMES runs its block exactly \_\_\_\_\_ times.
2. REPEAT UNTIL stops when its condition becomes \_\_\_\_\_.
3. An infinite loop's ending condition \_\_\_\_\_ becomes true.
4. FOR EACH visits \_\_\_\_\_ element of a list in order.
5. A sum accumulator should start at \_\_\_\_\_.
6. A product accumulator should start at \_\_\_\_\_.
7. A \_\_\_\_\_ keeps track of how many times something happens.
8. The statements that repeat are the loop \_\_\_\_\_.

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

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Practice

## Part 3 · How Many Times?

| Loop                                                                          | Times the body runs |
|-------------------------------------------------------------------------------|---------------------|
| REPEAT 5 TIMES                                                                | _____               |
| REPEAT 3 TIMES<br>containing REPEAT 4<br>TIMES (inner body)                   | _____               |
| $i \leftarrow 1$ , REPEAT<br>UNTIL ( $i > 10$ ) { $i$<br>$\leftarrow i + 3$ } | _____               |
| $x \leftarrow 20$ , REPEAT<br>UNTIL ( $x < 5$ ) { $x$<br>$\leftarrow x / 2$ } | _____               |
| $k \leftarrow 0$ , REPEAT<br>UNTIL ( $k = 7$ ) { $k$<br>$\leftarrow k + 2$ }  | _____               |

## Part 4 · Trace Table

```
total ← 0
n ← 1
REPEAT UNTIL (n > 5)
{
  IF (n MOD 2 = 1)
  { total ← total + n }
  n ← n + 1
}
DISPLAY(total)
```

| n<br>(start<br>of<br>pass)   | 1     | 2     | 3     | 4     | 5     |
|------------------------------|-------|-------|-------|-------|-------|
| total<br>(end<br>of<br>pass) | _____ | _____ | _____ | _____ | _____ |

Displayed: \_\_\_\_\_. In words, it adds the \_\_\_\_\_ numbers from 1 to 5.

## Part 5 · Fix the Loop

This should display 10, 8, 6, 4, 2.

```
n ← 10
REPEAT UNTIL (n = 1)
{
  DISPLAY(n)
  n ← n - 2
}
```

Problem: \_\_\_\_\_

Fix: \_\_\_\_\_

## Part 6 · Write a Loop

- a) Write a FOR EACH loop that counts how many elements of `answers` equal "yes".
- b) Write a REPEAT n TIMES loop that displays 5, 15, 25, 35.

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**Think:**

For REPEAT UNTIL, ask: Does the condition's variable change inside the loop? Will it EVER make the condition true? If not, it's an infinite loop.

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## Exam Practice

### Part 7 · Exam Practice

#### ★ Warm-up

1. What is x after this code?

```
x ← 3
REPEAT 4 TIMES
{ x ← x + 3 }
```

- A. 7
- B. 12
- C. 15
- D. 18

#### ★★ On Level

2. Which code displays 1 2 3 4?

- A.  $i \leftarrow 0$ , REPEAT 4 TIMES { DISPLAY(i)  $i \leftarrow i + 1$  }
- B.  $i \leftarrow 1$ , REPEAT UNTIL ( $i > 4$ ) { DISPLAY(i)  $i \leftarrow i + 1$  }
- C.  $i \leftarrow 1$ , REPEAT UNTIL ( $i = 4$ ) { DISPLAY(i)  $i \leftarrow i + 1$  }
- D.  $i \leftarrow 1$ , REPEAT UNTIL ( $i > 4$ ) {  $i \leftarrow i + 1$  DISPLAY(i) }

#### ★★ On Level

3. What is s?

```
vals ← [2, 5, 8]
s ← 0
FOR EACH v IN vals
{ s ← s + v * 2 }
```

- A. 30
- B. 60
- C. 15
- D. 16

#### ★★★ Exam Hard

4. What is displayed?

```
p ← 1
REPEAT 4 TIMES
{ p ← p * 2 }
DISPLAY(p)
```

- A. 8
- B. 16
- C. 32
- D. 4

#### ★★★ Exam Hard

5. Which TWO loops never stop? (Select two.)

- A.  $i \leftarrow 0$ , REPEAT UNTIL ( $i \geq 5$ ) {  $i \leftarrow i + 2$  }
- B.  $i \leftarrow 0$ , REPEAT UNTIL ( $i = 5$ ) {  $i \leftarrow i + 2$  }
- C. REPEAT 100 TIMES { DISPLAY("hi") }
- D.  $i \leftarrow 10$ , REPEAT UNTIL ( $i < 0$ ) {  $i \leftarrow i + 1$  }

### Part 8 · Written Response

Explain how to translate the Python loop `while x < 10:` into AP pseudocode, and why the condition flips.

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